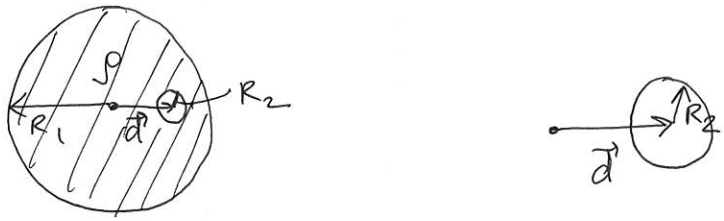
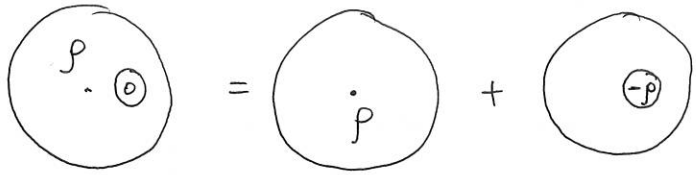
P1





$$0 = \rho + (-\rho)$$

Superposition principle: $\vec{E}_{cavity} = \vec{E}_{+\rho} + \vec{E}_{-\rho}$

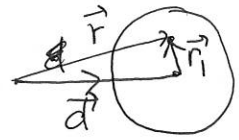
field of sphere R_1 with uniform density ρ

field of sphere R_2 with uniform density $(-\rho)$.

$$E_\rho(\vec{r}) \cdot 4\pi r^2 = \frac{\rho}{\varepsilon_0} \cdot \frac{4\pi r^3}{3}$$

$$E_\rho(r) = \frac{\rho r}{3\varepsilon_0} \Rightarrow \vec{E}_\rho(\vec{r}) = \frac{\rho \vec{r}}{3\varepsilon_0}$$ where $\vec{r}$ is measured from the center of the sphere

$$\vec{E}_{cavity}(\vec{r}) = \frac{\rho \vec{r}}{3\varepsilon_0} + \frac{(-\rho)}{3\varepsilon_0}\vec{r}_1 = \frac{\rho}{3\varepsilon_0}(\vec{r} - \vec{r}_1) = \frac{\rho}{3\varepsilon_0}\vec{d}$$



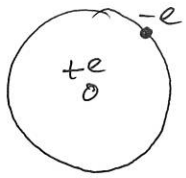
$\vec{d} + \vec{r}_1 = \vec{r}$

$\vec{r}_1 = \vec{r} - \vec{d}$

$\Rightarrow$ electric field inside empty cavity is <u>uniform</u>

$$\vec{E}_{cavity}(\vec{r}) = \frac{\rho \vec{d}}{3\varepsilon_0}.$$

P2



$$\rho_e(r) = \frac{-e}{\pi a^3} e^{-\frac{2r}{a}}, \quad \rho_p(r) = e\,\delta(\vec{r})$$

Use superposition principle: $\rho(r) = \rho_e + \rho_p$

$$\phi(\vec{r}) = \phi_e(\vec{r}) + \phi_p(\vec{r})$$

$\phi_p(\vec{r}) = \dfrac{e}{4\pi\varepsilon_0 r}$ potential of a point charge $+e$.

Find $\phi_e(\vec{r})$:

Gauss Theorem: $E_e(r)\cdot 4\pi r^2 = \underbrace{\int_0^r \frac{\rho_e}{\varepsilon_0} 4\pi r^2 dr}_{\text{charge enclosed within sphere of radius } r.}$

$$E_e(r) = \frac{1}{\varepsilon_0 r^2}\left(\frac{-e}{\pi a^3}\right)\int_0^r dr\, r^2 e^{-2r/a} = \left(x = \frac{2r}{a}\right)$$

$$= \frac{1}{\varepsilon_0 r^2}\left(\frac{-e}{\pi a^3}\right)\left(\frac{a}{2}\right)^3 \int_0^{2r/a} dx\, x^2 e^{-x} =$$

$$= \frac{1}{\varepsilon_0}\left(\frac{+e}{8\pi}\right)\frac{1}{r^2}\left\{e^{-2r/a}\left[2 + \frac{4r}{a} + \left(\frac{2r}{a}\right)^2\right] - 2\right\}$$

and $\vec{E}_e(\vec{r}) = E(r)\,\hat{r}$.

$\vec{E}_p(\vec{r}) = \dfrac{e}{4\pi\varepsilon_0 r^2}\,\hat{r}$.

the total electric field

$$\Rightarrow \vec{E}(r) = \vec{E}_e(r) + \vec{E}_p(r) = \frac{e}{8\pi\varepsilon_0}\cdot\frac{1}{r^2}\cdot e^{-2r/a}\left[2 + \frac{4r}{a} + \left(\frac{2r}{a}\right)^2\right]. \quad (1)$$

Next $\phi_e(\vec{R}) = \phi_e(R) = -\int_\infty^R dr\, E_e(r)$ since $\phi_e(R) - \phi_e(\infty) = +\int_\infty^R \underbrace{\nabla\phi}_{\frac{\partial\phi}{\partial r}} dr$ and $\phi_e(\infty) = 0$.

$$\phi_p(\vec{R}) = -\int_\infty^R dr\, E_p(r)$$

total $\phi(\vec{R}) = \phi_e(\vec{R}) + \phi_p(\vec{R}) = -\int_\infty^R dr\, E(r)$.

$$\phi(\vec{R}) = \int_R^\infty dr \left(\frac{e}{8\pi\varepsilon_0}\right) \frac{e^{-2r/a}}{r^2}\left(2 + \frac{4r}{a} + \left(\frac{2r}{a}\right)^2\right) =$$

$$= \frac{e}{4\pi\varepsilon_0} \int_R^\infty d\left(-\frac{1}{r}\right) e^{-2r/a} + \frac{e}{2\pi\varepsilon_0 a} \int_R^\infty dr \, \frac{1}{r} e^{-2r/a} +$$

$$+ \frac{e}{8\pi\varepsilon_0}\left(\frac{2}{a}\right)^2 \int_R^\infty dr \, e^{-2r/a} = \left(\frac{e}{4\pi\varepsilon_0}\right)\left\{ -\frac{e^{-2r/a}}{r}\Big|_R^\infty + \int_R^\infty d\left(e^{-2r/a}\right)\frac{1}{r}\right\}$$

$$+ \frac{e}{2\pi\varepsilon_0 a}\int_R^\infty dr \, \frac{e^{-2r/a}}{r} + \frac{e}{8\pi\varepsilon_0}\cdot\frac{2}{a}\int_{2R/a}^\infty dx \, e^{-x} =$$

$$= \frac{e}{4\pi\varepsilon_0}\left\{\frac{e^{-2R/a}}{R} - \frac{2}{a}\int_R^\infty dr \, \frac{e^{-2r/a}}{r}\right\} + \frac{e}{2\pi\varepsilon_0 a}\int_R^\infty dr \, \frac{e^{-2r/a}}{r}$$

$$+ \frac{e}{4\pi\varepsilon_0 a} e^{-2R/a} = \frac{e}{4\pi\varepsilon_0}\left(\frac{1}{R} + \frac{1}{a}\right) e^{-2R/a} \quad (2)$$

Check $\vec{E}(\vec{R}) = -\frac{\partial\phi}{\partial R}\hat{R} = \frac{e}{4\pi\varepsilon_0} e^{-2R/a}\left(\frac{1}{R^2} + \frac{2R}{a} + \frac{2}{a^2}\right)$ ✓

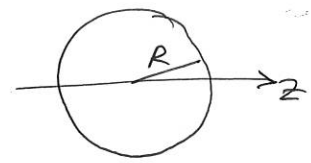
which agrees with Eq. (1) above.

Resultant electric field $\vec{E}(\vec{R})$ points outward because for any finite R the total charge enclosed inside sphere of radius R is positive – exponentially small tail of electron cloud extends to $R = \infty$.

(P3) Non-Uniformly charged sphere, $\rho(\theta) = \rho\cos\theta$.

$\Rightarrow$ Laplace (Poisson) equation for scalar potential ϕ for $r > R$ $(r < R)$

$$\nabla^2 \phi(\vec{r}) = -\frac{\rho\cos\theta}{\varepsilon_0} \quad \text{for } r < R.$$



$$\nabla^2 \phi(\vec{r}) = 0 \quad \text{for } r > R.$$

Problem is symmetric in φ angle $\Rightarrow \phi(\vec{r}) = \phi(r,\theta)$.

$$\nabla^2 \phi = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial \phi}{\partial r}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial \phi}{\partial \theta}\right)$$

let $\phi(r,\theta) = F_\nu(r)\cos\theta$. For $r < R$ $\nu = 1$:

$r<R$
$$\frac{\cos\theta}{r^2}\frac{d}{d}\left(r^2 \frac{dF_1(r)}{dr}\right) + \frac{F_1(r)}{r^2 \sin\theta}\underbrace{\frac{d}{d\theta}\left(\sin\theta \frac{d\cos\theta}{d\theta}\right)}_{\frac{d}{d\theta}\left(\sin\theta(-\sin\theta)\right) = -\frac{d}{d\theta}(\sin^2\theta) = -2\sin\theta\cos\theta} = -\frac{\rho\cos\theta}{\varepsilon_0}$$

$\Rightarrow$ obtain
$$\frac{\cos\theta}{r^2}\frac{d}{dr}\left(r^2 \frac{dF_1}{dr}\right) + \underbrace{\frac{F_1}{r^2 \sin\theta}\left(-2\sin\theta\cos\theta\right)}_{-\frac{F_1}{r^2}\cdot 2\cos\theta} = -\frac{\rho\cos\theta}{\varepsilon_0}$$

Can divide $(\cos\theta)$ out and multiply both sides by r^2:

$$(1) \quad \frac{d}{dr}\left(r^2 \frac{dF_1}{dr}\right) - 2F_1 = -\frac{\rho r^2}{\varepsilon_0} \quad (r < R).$$

For $r > R$ we then get

$$(2) \quad \frac{d}{dr}\left(r^2 \frac{dF_2}{dr}\right) - 2F_2 = 0 \quad (r > R)$$

(2) let $F_2 = r^\nu \Rightarrow \frac{d}{dr}\left(r^2 \nu r^{\nu-1}\right) = 2r^\nu$

$$\nu(\nu+1)\, r^\nu = 2r^\nu \Rightarrow \nu^2+\nu-2=0$$

$$\nu = \frac{1}{2}\left(-1 \pm \sqrt{1-4(-2)}\right) = \frac{1}{2}(-1\pm 3) = \begin{cases} 1 \\ -2 \end{cases}$$

$$\Rightarrow F_2 = A_2 r + \frac{B_2}{r^2}. \quad (r>R) \quad \text{(This is just } l=1 \text{ solution)}$$

Now Eq. (1) is nonhomogeneous $\Rightarrow F_1 = F_1^{(hom)} + F_1^{(par)}$

where $F_1^{(par)}$ is particular solution of non-homogeneous Eq. 1 while $F_1^{(hom)}$ is solution of the homogeneous equation (which is obtained by setting $\frac{\rho r^2}{\varepsilon_0} \to 0$ in its right-hand side).

Search for $F_1^{(par)}$: $F_1^{(par)} = \alpha r^2$

$$\Rightarrow \alpha \frac{d}{dr}\left(r^2 \cdot 2r\right) - 2\alpha r^2 = -\frac{\rho}{\varepsilon_0} r^2$$

$$2\alpha \cdot 3\, r^2 - 2\alpha r^2 = -\frac{\rho}{\varepsilon_0} r^2 \Rightarrow 4\alpha = -\frac{\rho}{\varepsilon_0}.$$

$$F_1^{(par)} = -\frac{\rho}{4\varepsilon_0} r^2.$$

And $F_1^{(hom)} = A_1 r + \frac{B_1}{r^2}$

$$\Rightarrow F_1(r) = F_1^{(hom)} + F_1^{(par)} = A_1 r + \frac{B_1}{r^2} - \frac{\rho}{4\varepsilon_0} r^2. \quad (r<R)$$

Boundary conditions at $r=0$: $F_1(r)$ must be finite $\Rightarrow B_1 = 0$.

at $r=\infty$: $F_2(r) = 0 \Rightarrow A_2 = 0$.

$$\Rightarrow \begin{cases} F_1 = \cancel{\frac{B_1}{r^2}} - \frac{\rho r^2}{4\varepsilon_0} + A_1 r \\ F_2 = \frac{B_2}{r^2} \end{cases}$$

Boundary condition at $r=R$: $\phi_1(R,\theta) = \phi_2(R,\theta)$,

$$E_{1n} = E_{2n} \Rightarrow \frac{\partial \phi_1(R,\theta)}{\partial n} = \frac{\partial \phi_2(R,\theta)}{\partial n}$$ since there is no surface charge density

$$\begin{cases} A_1 R - \dfrac{\rho R^2}{4\varepsilon_0} = \dfrac{B_2}{R^2} \\ A_1 - \dfrac{\rho}{2\varepsilon_0} R = -\dfrac{2B_2}{R^3}, \quad A_1 R - \dfrac{\rho}{2\varepsilon_0} R^2 = -\dfrac{2B_2}{R^2} \end{cases}$$

$$\Rightarrow \underbrace{-\frac{\rho R^2}{4\varepsilon_0} - \left(-\frac{\rho R^2}{2\varepsilon_0}\right)}_{} = \frac{B_2}{R^2}\left(1-(-2)\right) = \frac{3B_2}{R^2}$$

$$+\frac{\rho R^2}{4\varepsilon_0} = \frac{3B_2}{R^2} \Rightarrow B_2 = +\frac{\rho R^4}{12\varepsilon_0}.$$

$$\Rightarrow A_1 = \frac{\rho R}{2\varepsilon_0} - \frac{2}{R^3}\left(+\frac{\rho R^4}{12\varepsilon_0}\right) = \frac{\rho R}{2\varepsilon_0} - \frac{\rho R}{6\varepsilon_0} = \frac{\rho R}{6\varepsilon_0} = \frac{\rho R}{3\varepsilon_0}$$

$$F_1 = A_1 r - \frac{\rho r^2}{4\varepsilon_0} = \frac{\rho R}{3\varepsilon_0} r - \frac{\rho r^2}{4\varepsilon_0} = \frac{\rho}{4\varepsilon_0} r\left(\frac{4}{3}R - r\right)$$

$$\phi_1(r,\theta) = \frac{\rho}{4\varepsilon_0} r\left(\frac{4}{3}R - r\right)\cos\theta; \; r<R.$$

$$F_2 = \frac{B_2}{r^2} = \frac{\rho R^4}{12\varepsilon_0}\frac{1}{r^2} \Rightarrow \phi_2(r,\theta) = \frac{\rho R^4}{12\varepsilon_0}\frac{\cos\theta}{r^2}, \; r>R$$

$$\vec{E}_\nu = -\vec{\nabla}\phi_\nu = -\left(\frac{\partial \phi_\nu}{\partial r}\hat{r} + \frac{1}{r}\frac{\partial \phi_\nu}{\partial \theta}\hat{\theta}\right)$$

$$\vec{E}_1(r,\theta) = -\left\{\left(\frac{\rho}{3\varepsilon_0}R - \frac{\rho}{2\varepsilon_0}r\right)\cos\theta\,\hat{r} - \frac{\rho}{4\varepsilon_0}\left(\frac{4R}{3} - r\right)\sin\theta\,\hat{\theta}\right\}.$$

$$\vec{E}_2(r,\theta) = -\left\{-\frac{\rho R^4}{6\varepsilon_0}\frac{\cos\theta}{r^3}\hat{r} - \frac{\rho R^4}{12\varepsilon_0}\frac{\sin\theta}{r^3}\hat{\theta}\right\}$$

At $r>R$ this is field of electric dipole.